FIT5201 - Machine Learning

Assignment 1 :

- Interviews in Weeks 8-9
- Choose interview slot offered by allocated interviewer (details posted next week)
- Must show up to interview (otherwise zero marks for whole assignment)

Module 3:

Linear Models for Classification

(Part B)

Module Objectives

Understand

- Basic Concepts of Classification
- Discriminative Models / Geometric Perspective
- Extension to Multiple Classes
- Historic Note: Perceptron Algorithm
- Probabilistic Model: Logistic Regression
- Generative Models: Bayes Classifier (Single Input)
- Handling Multiple Inputs: Naïve Bayes and Multivariate Gaussian

Logistic Regression

Classification problem

Given

- data $D = (x_1, t_1), \dots, (x_N, t_N)$
- sampled according to joint distribution of RVs X and T
- input variable(s) X often assumed with $D_X = \mathbb{R}^d$
- discrete target variable T with $D_T = \{c_1, \dots, c_K\}$

Goal

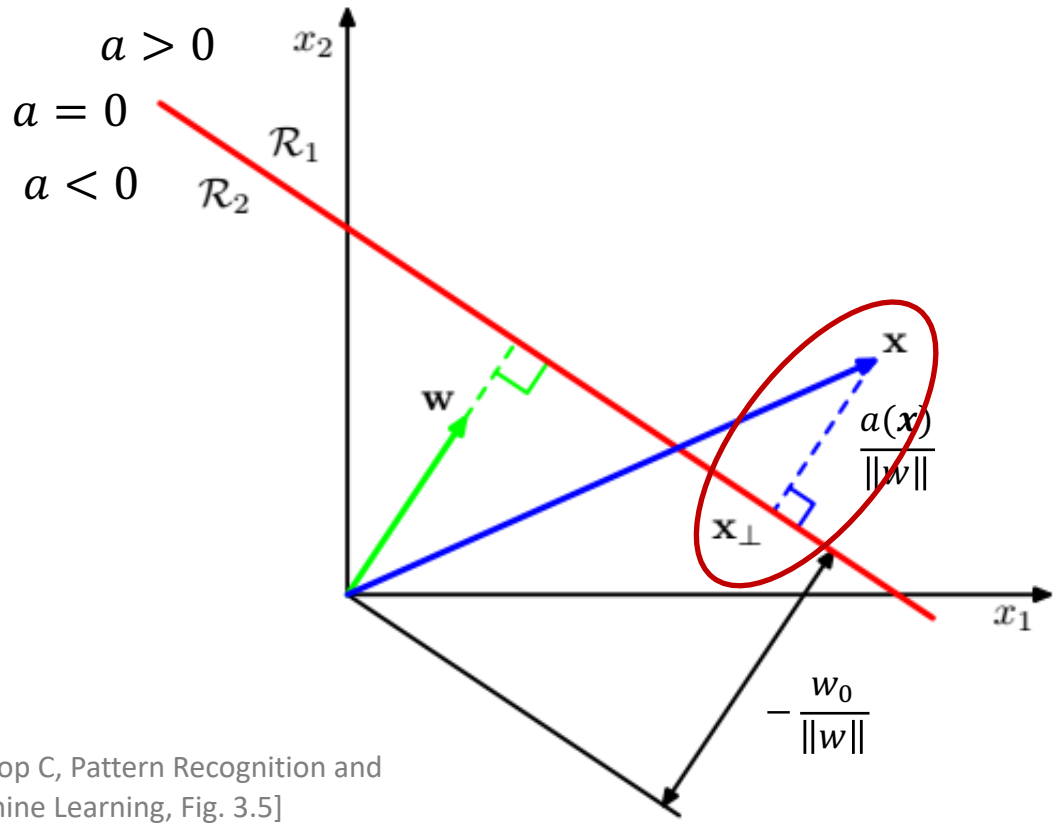
- learn parameters $\boldsymbol{w}$ of prediction function $y(\boldsymbol{x}, \boldsymbol{w})$
- that minimise prediction risk / (population) test error / generalisation error

$$\boldsymbol{E}\big[l\big(T, y(X, \boldsymbol{w})\big)\big]$$

- …with some loss function l such as 0/1-loss

$$l(t, y) = \begin{cases} 0, & \text{if } t = y \\ 1, & \text{otherwise} \end{cases}$$

Discriminative Models



[Bishop C, Pattern Recognition and Machine Learning, Fig. 3.5]

Decision boundary: $y(\boldsymbol{x}) = \boldsymbol{w}^T\boldsymbol{x} = 0$

If $y(\boldsymbol{x}_1) > y(\boldsymbol{x}_2) > 0$ then, relatively, more certain that $t_1 = 1$ than $t_2 = 1$.

Problem:
What does $y(\boldsymbol{x}_1)$ mean in absolute terms?

Would like probabilistic model:

$$p(T = 1|\boldsymbol{X} = \boldsymbol{x})$$

Logit and Sigmoid Transforms

Problem: linear function maps to whole real line, hence can not directly represent probability

Idea: let linear function represent the log odds:

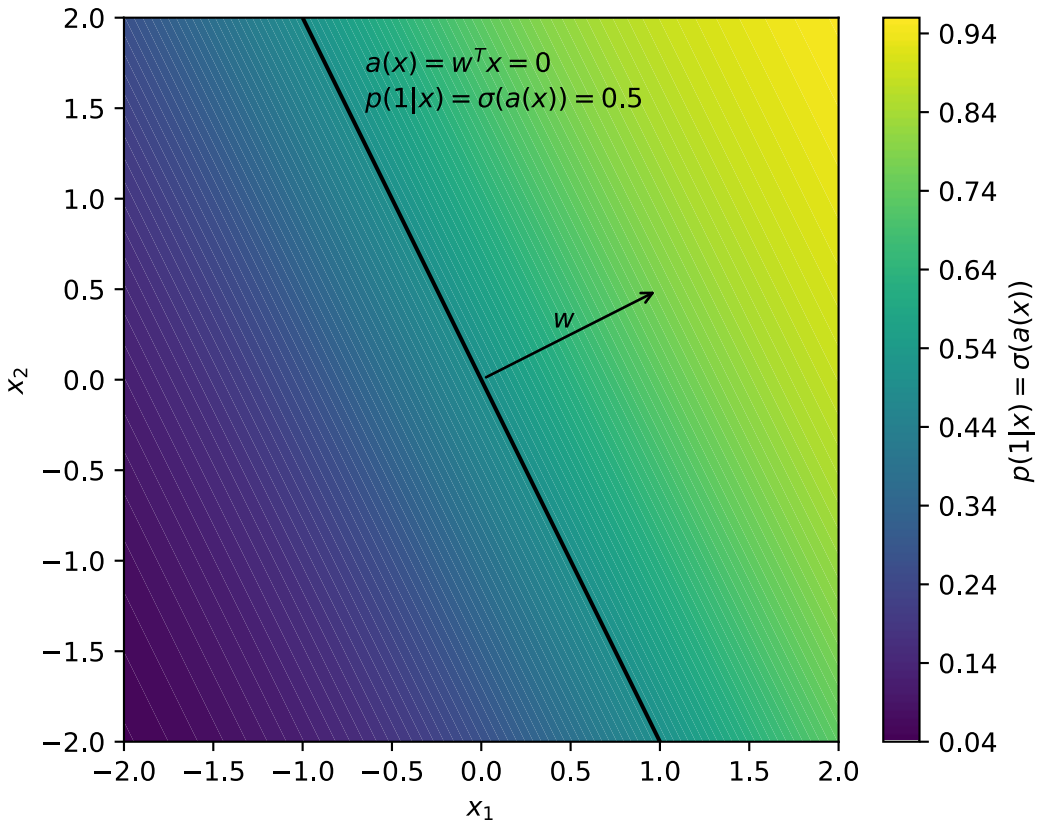
$$a(x) = \boldsymbol{w}^T \boldsymbol{x} = \ln \frac{p(T = 1|\boldsymbol{X} = \boldsymbol{x})}{p(T = 0|\boldsymbol{X} = \boldsymbol{x})}$$

$$y(x) = p(T = 1|\boldsymbol{X} = \boldsymbol{x}) = \sigma(a(\boldsymbol{x}))$$

Sigmoid function "unpacks" probability from log odds.

$$\sigma(a) = \frac{1}{1 + \exp(-a)}$$

$$= \frac{1}{1 + \frac{p(0|\boldsymbol{x})}{p(1|\boldsymbol{x})}} = \frac{p(1|\boldsymbol{x})}{p(1|x) + p(0|\boldsymbol{x})} = p(1|\boldsymbol{x})$$



Sigmoid Function

Definition

$$\sigma(a) = \frac{1}{1 + \exp(-a)}$$

Properties

$$\sigma(-a) = 1 - \sigma(a)$$

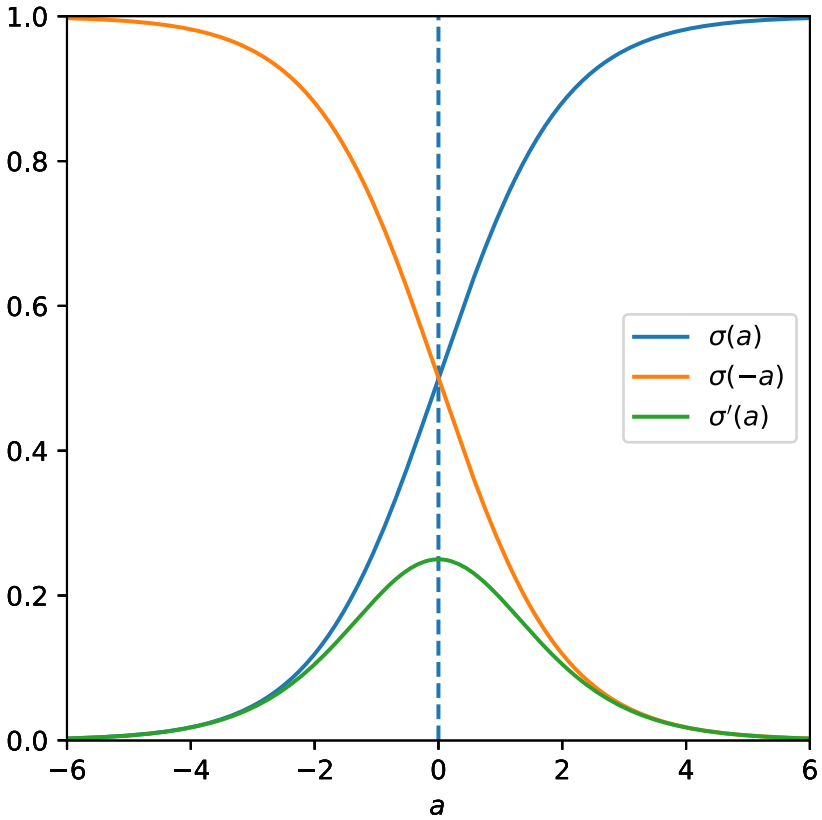
clear symmetry, as

$$-a(x) = \ln \frac{p(0|\boldsymbol{x})}{p(1|\boldsymbol{x})}$$

switches role of classes

$$\frac{\partial}{\partial a}\sigma(a) = \sigma(a)\sigma(-a)$$

Probability varies most close to decision boundary



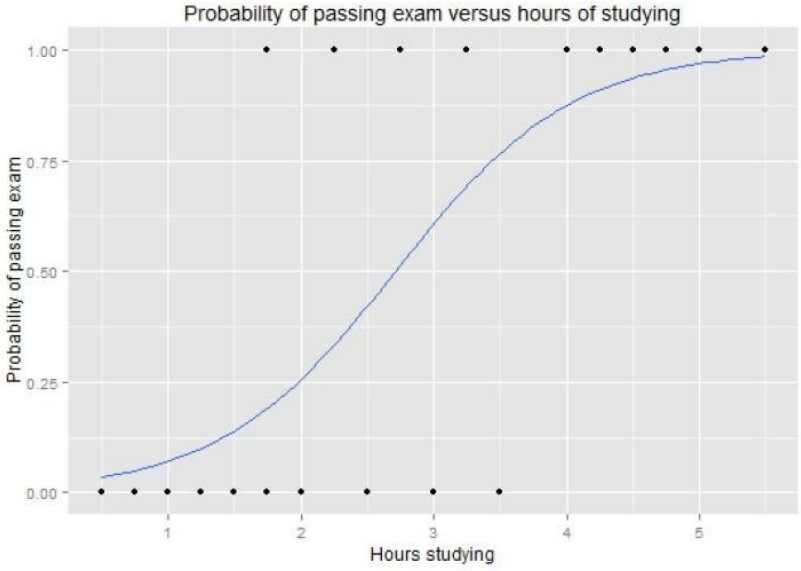
View as Generalised Linear Model

Predict probability / expected value of T:

$$y(\boldsymbol{x}) = \sigma(\boldsymbol{w}^T\boldsymbol{x} + w_0) = \mathbf{E}[T|X = \boldsymbol{x}]$$

Example: probability of passing an exam ($T = 1$) versus hours of study (X)

Hours	0.50	0.75	1.00	1.25	1.50	1.75	1.75	2.00	2.25	2.50	2.75	3.00	3.25	3.50	4.00	4.25	4.50	4.75	5.00	5.50
Pass	0	0	0	0	0	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1



Hours of study	Probability of passing exam
1	0.07
2	0.26
3	0.61
4	0.87
5	0.97

$$y(x) = \sigma(w_1 x + w_0)$$
$$= \sigma(1.5x - 4.078)$$

Task: learn parameters $\boldsymbol{w}$

Logistic Regression

Use maximum likelihood to learn model parameters $\boldsymbol{w}$ from iid data

$$D = \{(x_1, t_1), \dots, (x_N, t_N)\}$$

With convention $t_n = 1$ (positive) and $t_n = 0$ (negative) and $y_n = p(1|\boldsymbol{x}_n) = \sigma(\boldsymbol{w}^T \boldsymbol{x}_n)$ can write negative log likelihood function as

$$\begin{aligned}
-L(\boldsymbol{w}) &= -\ln \prod_{n=1}^{N} p(t_n|\boldsymbol{x}_n, \boldsymbol{w}) \\
&= -\ln \prod_{n=1}^{N} y_n^{t_n} (1 - y_n)^{1-t_n} \\
&= -\sum_{n=1}^{N} t_n \ln y_n + (1 - t_n) \ln(1 - y_n) \\
&= \sum_{n=1}^{N} l_{\log}(t_n, y_n)
\end{aligned}$$

tricky way to write case distinction

$$\begin{cases} y_n, & t_n = 1 \\ 1 - y_n, & t_n = 0 \end{cases}$$

Gives rise to logistic loss:

$$l_{\log}(t, y) = \begin{cases} -\ln y\,, & t = 1 \\ -\ln(1 - y)\,, & t = 0 \end{cases}$$

Logistic Regression

To minimise corresponding error function

$$E(\boldsymbol{w}) = -\sum_{n=1}^{N} t_n \ln y_n + (1 - t_n)\ln(1 - y_n)$$

$$= -\sum_{n=1}^{N} t_n \ln \sigma(\boldsymbol{w}^T \boldsymbol{x}_n) + (1 - t_n)\ln \sigma(-\boldsymbol{w}^T \boldsymbol{x}_n)$$

find its gradient

$$\nabla E(\boldsymbol{w}) = -\sum_{n=1}^{N} t_n \nabla \ln \sigma(\boldsymbol{w}^T \boldsymbol{x}_n) + (1 - t_n)\nabla \ln \sigma(-\boldsymbol{w}^T \boldsymbol{x}_n)$$

$$= -\sum_{n=1}^{N} t_n(1 - y_n)\boldsymbol{x}_n - (1 - t_n)y_n \boldsymbol{x}_n$$

$$= -\sum_{n=1}^{N} (t_n - y_n)\, \boldsymbol{x}_n$$

Using chain rule and properties of σ:

$$\frac{\partial}{\partial w_i} \ln \sigma(\boldsymbol{w}^T \boldsymbol{x}_n)$$

$$= \frac{1}{\sigma(\boldsymbol{w}^T \boldsymbol{x}_n)} \frac{\partial}{\partial w_i} \sigma(\boldsymbol{w}^T \boldsymbol{x}_n)$$

$$= \frac{\sigma(\boldsymbol{w}^T \boldsymbol{x}_n)\sigma(-\boldsymbol{w}^T \boldsymbol{x}_n)}{\sigma(\boldsymbol{w}^T \boldsymbol{x}_n)} \frac{\partial}{\partial w_i} \boldsymbol{w}^T \boldsymbol{x}_n$$

$$= \sigma(-\boldsymbol{w}^T \boldsymbol{x}_n) x_{n,i}$$

$$\Downarrow$$

$$\nabla \ln \sigma(\boldsymbol{w}^T \boldsymbol{x}_n) = \sigma(-\boldsymbol{w}^T \boldsymbol{x}_n)\boldsymbol{x}_n$$

$$= (1 - y_n)\boldsymbol{x}_n$$

Gradient Descent

$$\nabla E(\boldsymbol{w}) = -\sum_{n=1}^{N} (t_n - y_n)\, \boldsymbol{x}_n$$

$$= \boldsymbol{X}^T(\boldsymbol{y} - \boldsymbol{t})$$

$$= \boldsymbol{X}^T(\sigma(\boldsymbol{X}\boldsymbol{w}) - \boldsymbol{t})$$

Similar to gradient in (least squares) linear regression:

$$\boldsymbol{X}^T(\boldsymbol{X}\boldsymbol{w} - \boldsymbol{t})$$

"vectorised" form

$$\boldsymbol{X} = \begin{bmatrix} \boldsymbol{x}_1 \\ \boldsymbol{x}_2 \\ \vdots \\ \boldsymbol{x}_N \end{bmatrix} \qquad \boldsymbol{t} = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_N \end{bmatrix} \qquad \boldsymbol{y} = \begin{bmatrix} \sigma(\boldsymbol{x}_1^T\boldsymbol{w}) \\ \sigma(\boldsymbol{x}_2^T\boldsymbol{w}) \\ \vdots \\ \sigma(\boldsymbol{x}_N^T\boldsymbol{w}) \end{bmatrix} = \sigma(\boldsymbol{X}\boldsymbol{w})$$

input matrix target vector prediction vector

Stochastic Gradient Update

$$\boldsymbol{w}_\tau = \boldsymbol{w}_{\tau-1} - \eta\tilde{\boldsymbol{X}}^T(\sigma(\tilde{\boldsymbol{X}}\boldsymbol{w}_{\tau-1}) - \tilde{\boldsymbol{t}})$$

where $\tilde{\boldsymbol{X}}$ and $\tilde{\boldsymbol{t}}$ contain rows of $\boldsymbol{X}$ and $\boldsymbol{t}$ in batch of iteration τ

Logistic Regression: Closing Remarks

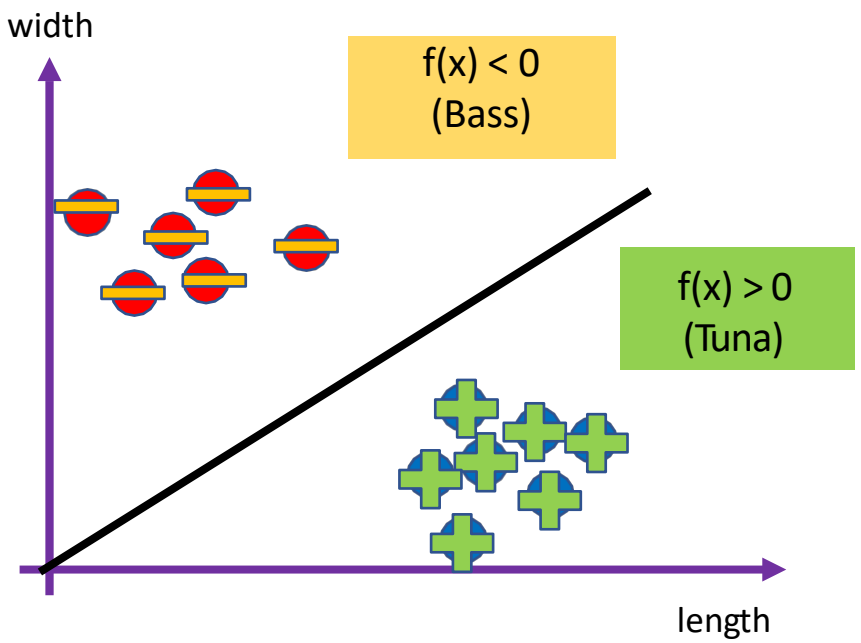
- Logistic regression uses linear function to model log odds of two classes (or one versus another)
- Parameterising log odds is widely used for probabilistic classification (not only in linear models)
- Interpretable model: input weight magnitude indicates importance of corresponding input variable
- **Warning**: (stochastic) gradient descent does not converge for linearly separable data; reason: will push norm of weights to infinity to approach $p(t_n|\boldsymbol{x}_n) = 1$ for all examples
- One generally uses regularisation to avoid this
- Generalisation to multiple classes possible (uses variant of sigmoid called "soft max")

Probabilistic Generative Models

Discriminative vs Generative Classifiers

Discriminative classifiers

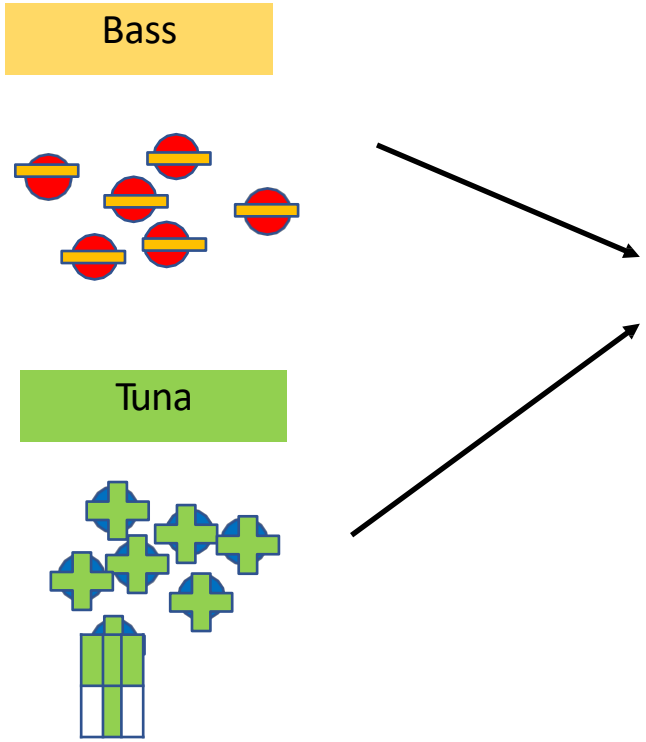
- **Approach 1:** Search for a decision boundary that separates class labels.



Discriminative vs Generative Classifiers

Probabilistic classifiers

- **Approach 2:** Try to learn $p(c_k|\boldsymbol{x})$ directly

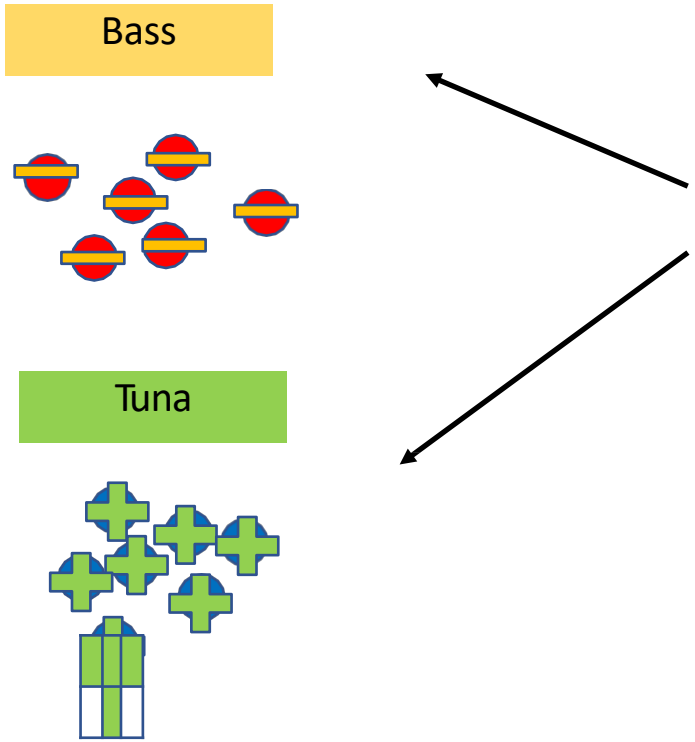


Learn Class Probability:
$p(c_k|\boldsymbol{x})$

Discriminative vs Generative Classifiers

Generative classifiers

- Try to model $p(\boldsymbol{x}|c_k)$, i.e., how do inputs look like for class k
- this can actually be used to **generate** new input data



Model:

- $p(\mathbf{x}|\mathrm{T} = \mathrm{Tuna})$: model the distributions of features of Tuna
- $p(\mathbf{x}|\mathrm{T} = \mathrm{Bass})$: model the distributions of features of Bass

Probabilistic Generative Models

Idea: instead of modelling $p(c_k|\boldsymbol{x})$, model

- class conditional distribution of inputs $p(\boldsymbol{x}|c_k)$
- class prior (marginal) distribution $p(c_k)$

Then obtain required class probabilities via

$$p(c_k|\boldsymbol{x}) = \frac{p(\boldsymbol{x}|c_k)p(c_k)}{p(\boldsymbol{x})} = \frac{p(\boldsymbol{x}|c_k)p(c_k)}{\sum_{k=1}^{K} p(\boldsymbol{x}|c_k)p(c_k)}$$

Why?

- Model of full joint distribution allows for imputation of missing values, generation of alternative inputs (e.g., in AI art or “deep fakes”).
- Class conditionals often straightforward to model/learn.

DALL-E 2 generated this image when given the prompt "Teddy bears working on new AI research underwater with 1990s technology".

Class Prior Distribution

The class prior distribution is a simple discrete categorical (or multinomial) distribution given by K probabilities

$$\boldsymbol{\varphi} = (\varphi_1, \dots, \varphi_K)$$

that add up to 1:

$$\varphi_1 + \varphi_1 + \cdots + \varphi_K = 1$$

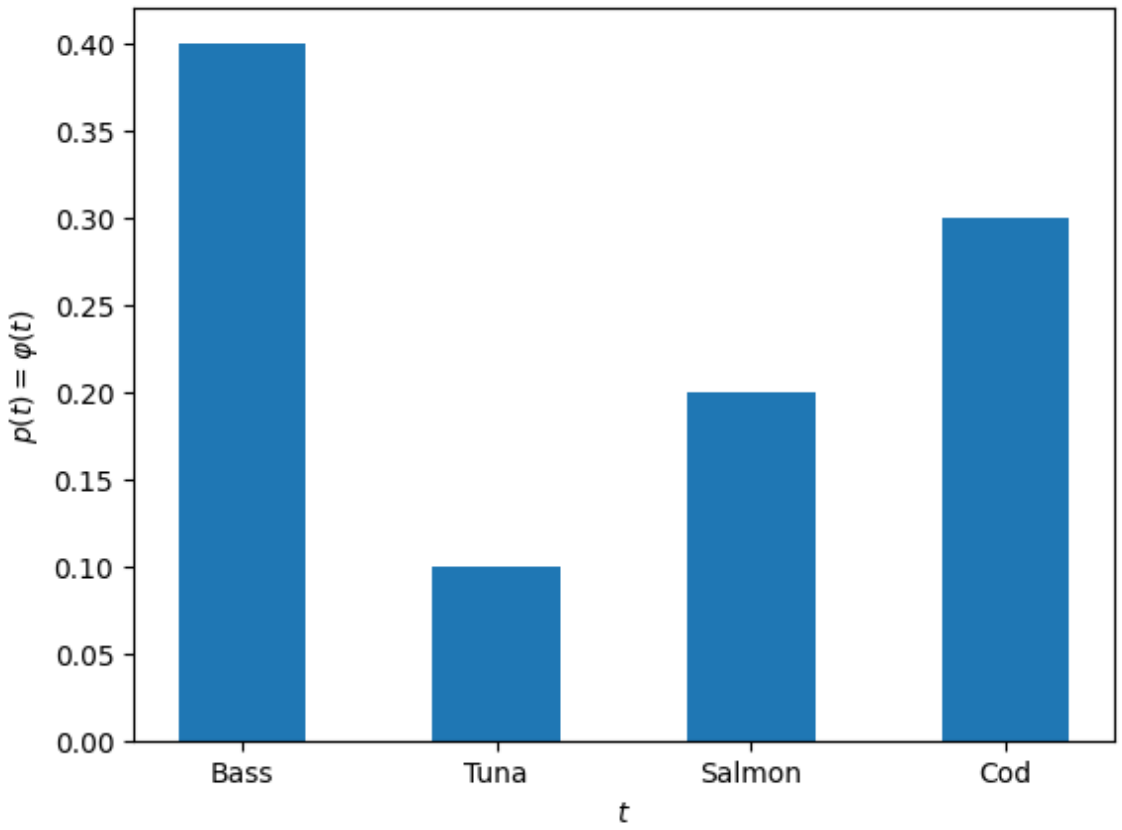
That is, the mass function is simply:

$$p(c_k|\boldsymbol{\varphi}) = \varphi_k$$

The maximum likelihood estimate is given by counting occurrence of classes:

$$\varphi_k = \frac{\boxed{N_k}}{N}$$

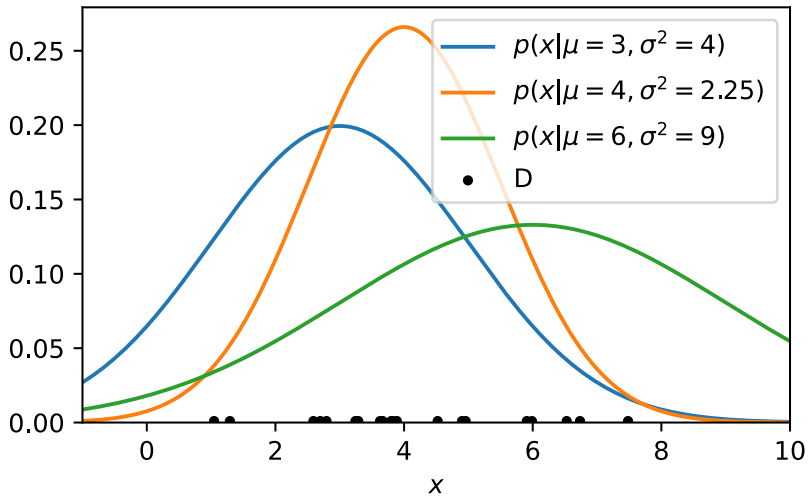
number of data points of class k



Class Conditional Densities

For single continuous input variable X, can assume that normal class conditional:

$$p(x|c_k) = \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{(x-\mu_k)^2}{2\sigma_k^2}\right)$$



Need to fit parameters μ_k and σ_k for each class.

Split training data

$$D = \{(x_1, t_1), \dots, (x_N, t_N)\}$$

into K parts, one for each class

$$D_k = \{(x_n, t_n) : n \in I_k\}$$
$$I_k = \{n : t_n = k\}$$

Fitting Parameters for Class Conditionals

Likelihood function

$$p(D_k|\mu_k,\sigma_k) = \prod_{n\in I_k} \frac{1}{\sqrt{2\pi}\sigma_k} \exp\left(-\frac{(x_n-\mu_k)^2}{2\sigma_k^2}\right)$$

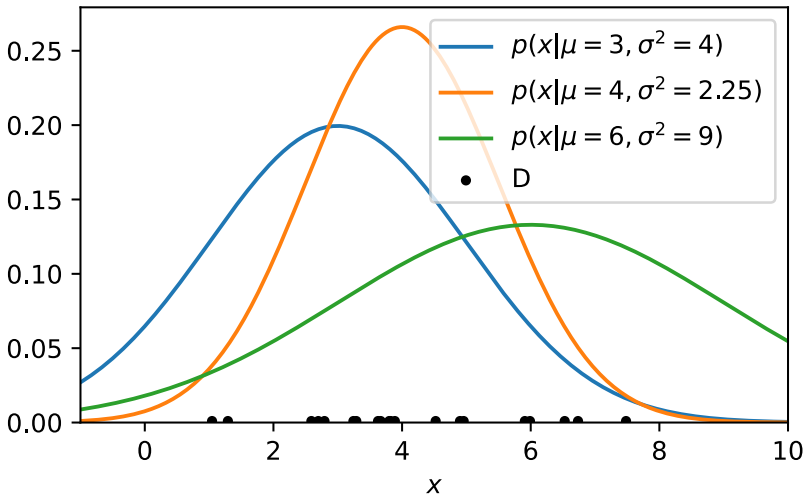
Log likelihood

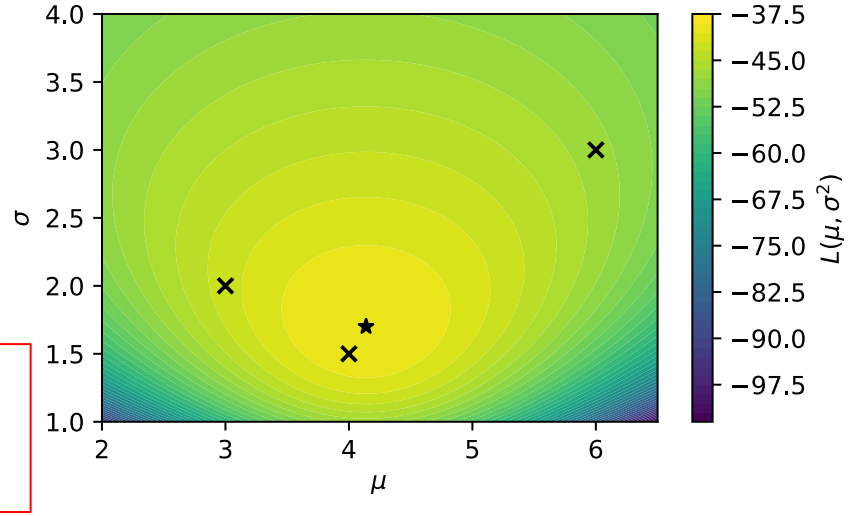
$$L(\mu_k,\sigma_k) = -\sum_{n\in I_k} \ln\sqrt{2\pi}\sigma_k - \sum_{n\in I_k} \frac{(x_n-\mu_k)^2}{2\sigma_k^2}$$

Derivative criterion

$$\frac{\partial L(\mu_k,\sigma_k)}{\partial \mu_k} = \sum_{n\in I_k} \frac{x_n-\mu_k}{\sigma_k^2} = 0 \qquad \boxed{\Rightarrow \mu_k = \sum_{n\in I_k} x_n / N_k}$$

$$\frac{\partial L(\mu_k,\sigma_k)}{\partial \sigma_k} = -\frac{N_k}{\sigma_k} + \sum_{n\in I_k} \frac{(x_n-\mu_k)^2}{\sigma_k^3} = 0 \qquad \boxed{\Rightarrow \sigma_k^2 = \sum_{n\in I_k} (x_n-\mu_k)^2 / N_k}$$





How to Handle Multiple Input Variables

Assume again multiple (continuous) input variables: $\boldsymbol{X} = (X_1, \dots, X_d) \in \mathbb{R}^d$.

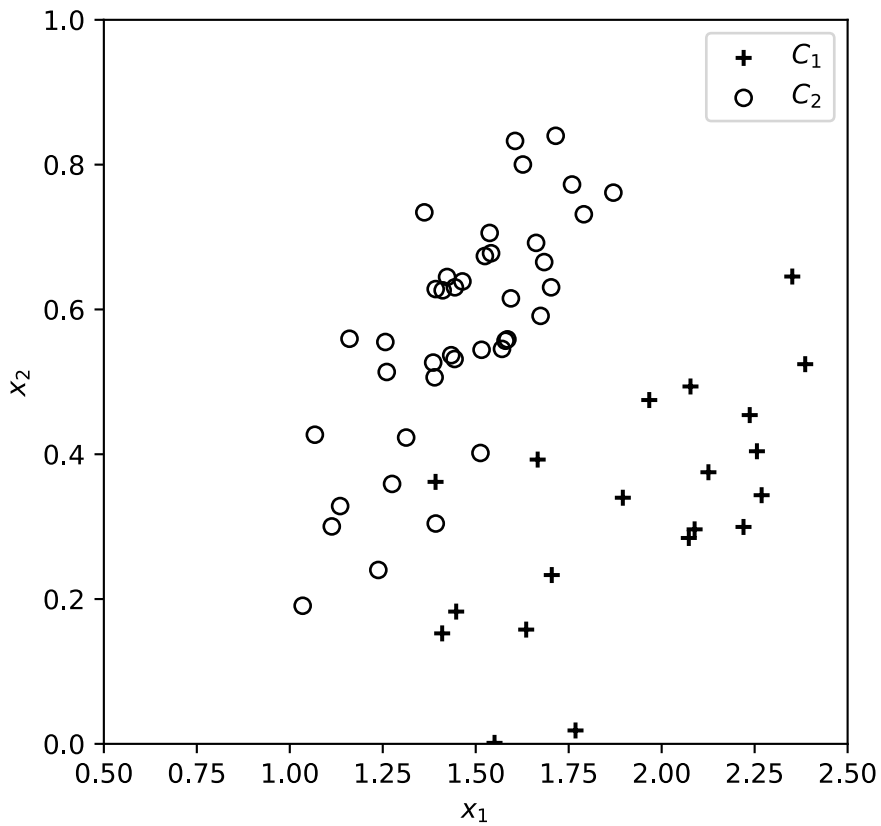
How to model class conditional input densities?

Idea 1: assume conditionally independent inputs

$$p(\boldsymbol{x}|c_k) = p(x_1|c_k)p(x_2|c_k) \cdots p(x_d|c_k)$$

-> allows to re-use previous univariate Gaussians
-> approach is called Naïve Bayes
-> however, independence assumption often violated

Idea 2: model as multivariate normal distribution



Covariance Matrices and the Multivariate Normal Distribution

Scenario: Mean and Variance of Grocery Stores Profits

Assume we have a grocery stores selling the following products:

Product	Profit	Sold	Mn. Sold $\mathbf{E}(X)$	Std. Sold $\sqrt{\mathbf{Var}(X)}$
apples	1	X_1	100	10
pears	2	X_2	50	10
cabbage	4	X_3	25	2

What is the expected value of the profit $Y = X_1 + 2X_2 + 4X_3$?

$$\mathbf{E}(Y) = \mathbf{E}(X_1 + 2X_2 + 4X_3)$$
$$= \mathbf{E}(X_1) + 2\mathbf{E}(X_2) + 4\mathbf{E}(X_3)$$
$$= 100 + 100 + 100 = 300$$

What is the variance / stan. dev. of the profit?

Can't tell without knowing covariances!!

Covariance

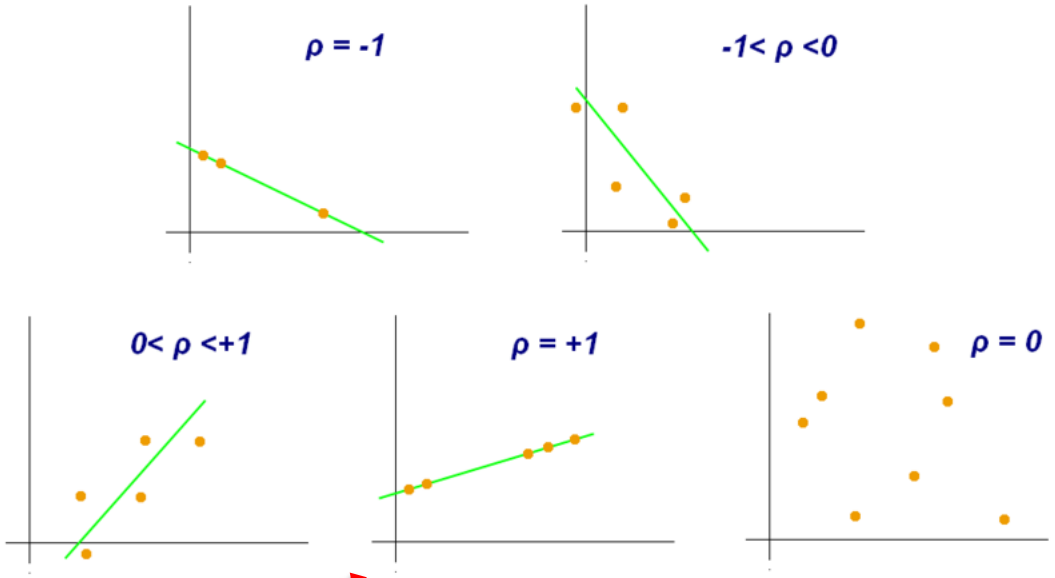
Covariance of two random variables X and Y with means μ and ν:

$$\mathbf{Cov}(X,Y) = \mathbf{E}\big((X-\mu)(Y-\nu)\big) = \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y)$$

determines variance of sums of random variables:

$$\mathbf{Var}(X+Y) = \mathbf{Var}(X) + \mathbf{Var}(Y) + 2\mathbf{Cov}(X,Y)$$

$$\rho(X,Y) = \frac{\mathbf{Cov}(X,Y)}{\sqrt{\mathbf{Var}(X)\mathbf{Var}(Y)}}$$



Properties:

- Generalises var.: $\mathbf{Var}(X) = \mathbf{Cov}(X,X)$
- Symmetric: $\mathbf{Cov}(X,Y) = \mathbf{Cov}(Y,X)$
- Shift invariant: $\mathbf{Cov}(X+c, Y+d) = \mathbf{Cov}(Y,X)$
- Bounded: $|\mathbf{Cov}(X,Y)| \leq \sqrt{\mathbf{Var}(X)\mathbf{Var}(Y)}$

gives rise to Pearson correlation coefficient (normalised covariance)

Covariance

Covariance of two random variables X and Y with means μ and ν:

$$\mathbf{Cov}(X,Y) = \mathbf{E}\big((X-\mu)(Y-\nu)\big) = \mathbf{E}(XY) - \mathbf{E}(X)\mathbf{E}(Y)$$

determines variance of sums of random variables:

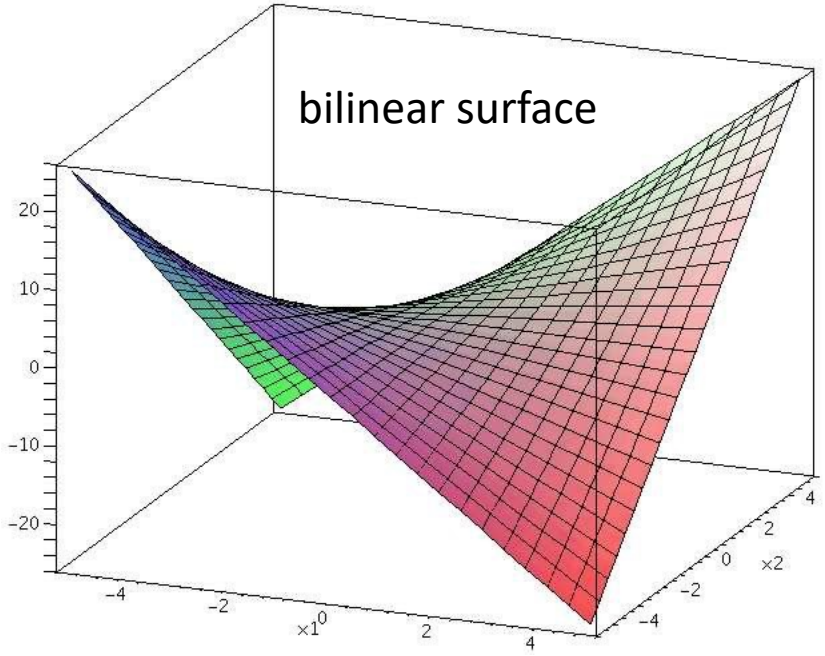
$$\mathbf{Var}(X+Y) = \mathbf{Var}(X) + \mathbf{Var}(Y) + 2\mathbf{Cov}(X,Y)$$

Bilinearity Proof

$$\begin{aligned}
&\mathbf{Cov}(aX+bY,Z)\\
&= \mathbf{E}\big((aX+bY)Z\big) - \mathbf{E}(aX+bY)\mathbf{E}(Z)\\
&= \mathbf{E}(aXZ+bYZ) - \big(\mathbf{E}(aX)+\mathbf{E}(bY)\big)\mathbf{E}(Z)\\
&= a\mathbf{E}(XZ) + b\mathbf{E}(YZ) - a\mathbf{E}(X)\mathbf{E}(Z) - b\mathbf{E}(Y)\mathbf{E}(Z)\\
&= a\mathbf{Cov}(X,Z) + b\mathbf{Cov}(Y,Z)
\end{aligned}$$

Properties:

- Generalises var.: $\mathbf{Var}(X) = \mathbf{Cov}(X,X)$
- Symmetric: $\mathbf{Cov}(X,Y) = \mathbf{Cov}(Y,X)$
- Shift invariant: $\mathbf{Cov}(X+c,Y+d) = \mathbf{Cov}(Y,X)$
- Bounded: $|\mathbf{Cov}(X,Y)| \le \sqrt{\mathbf{Var}(X)\mathbf{Var}(Y)}$
- Bilinear: $\mathbf{Cov}(aX+bY,Z) = a\mathbf{Cov}(X,Z) + b\mathbf{Cov}(Y,Z)$



Scenario: Mean and Variance of Grocery Stores Profits

Assume we have a grocery stores selling the following products:

Product	Profit	Sold	Mn. Sold $\mathbf{E}(X)$	Std. Sold $\sqrt{\mathbf{Var}(X)}$
apples	1	X_1	100	10
pears	2	X_2	50	10
cabbage	4	X_3	25	2

Cov	X_1	X_2	X_3
X_1	100	50	-10
X_2	50	100	-10
X_3	-10	-10	4

ρ	X_1	X_2	X_3
X_1	1	0.5	-0.5
X_2	0.5	1	-0.5
X_3	-0.5	-0.5	1

What is the variance of the profit $Y = X_1 + 2X_2 + 4X_3$?

$$\mathbf{Var}(Y_1) = \mathbf{Var}(X_1 + 2X_2 + 4X_3) = \mathbf{Cov}(X_1 + 2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

$$= \mathbf{Cov}(X_1, X_1 + 2X_2 + 4X_3) + \mathbf{Cov}(2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

$$= \mathbf{Cov}(X_1, X_1) + \mathbf{Cov}(X_1, 2X_2 + 4X_3) + \mathbf{Cov}(2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

$$= \mathbf{Cov}(X_1, X_1) + 2\mathbf{Cov}(X_1, X_2) + 4\mathbf{Cov}(X_1, X_3) + \mathbf{Cov}(2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

Scenario: Mean and Variance of Grocery Stores Profits

Assume we have a grocery stores selling the following products:

Product	Profit	Sold	Mn. Sold $\mathbf{E}(X)$	Std. Sold $\sqrt{\mathbf{Var}(X)}$
apples	1	X_1	100	10
pears	2	X_2	50	10
cabbage	4	X_3	25	2

Cov	X_1	X_2	X_3
X_1	100	50	-10
X_2	50	100	-10
X_3	-10	-10	4

ρ	X_1	X_2	X_3
X_1	1	0.5	-0.5
X_2	0.5	1	-0.5
X_3	-0.5	-0.5	1

What is the variance of the profit $Y = X_1 + 2X_2 + 4X_3$?

$$\mathbf{Var}(Y_1) = \mathbf{Var}(X_1 + 2X_2 + 4X_3) = \mathbf{Cov}(X_1 + 2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

...

$$= \mathbf{Cov}(X_1, X_1) + 2\mathbf{Cov}(X_1, X_2) + 4\mathbf{Cov}(X_1, X_3) + \mathbf{Cov}(2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

$$= \mathbf{Cov}(X_1, X_1) + 2\mathbf{Cov}(X_1, X_2) + 4\mathbf{Cov}(X_1, X_3) + 2\mathbf{Cov}(X_2, X_1 + 2X_2 + 4X_3) + 4\mathbf{Cov}(X_3, X_1 + 2X_2 + 4X_3)$$

$$= \mathbf{Cov}(X_1, X_1) + 2\mathbf{Cov}(X_1, X_2) + 4\mathbf{Cov}(X_1, X_3) + 2\mathbf{Cov}(X_2, X_1) + \mathbf{Cov}(2X_2 + 4X_3) + 4\mathbf{Cov}(X_3, X_1 + 2X_2 + 4X_3)$$

$$= \mathbf{Cov}(X_1, X_1) + 4\mathbf{Cov}(X_1, X_2) + 4\mathbf{Cov}(X_1, X_3) + \mathbf{Cov}(2X_2 + 4X_3) + 4\mathbf{Cov}(X_3, X_1 + 2X_2 + 4X_3)$$

Scenario: Mean and Variance of Grocery Stores Profits

Assume we have a grocery stores selling the following products:

Product	Profit	Sold	Mn. Sold $\mathbf{E}(X)$	Std. Sold $\sqrt{\mathbf{Var}(X)}$
apples	1	X_1	100	10
pears	2	X_2	50	10
cabbage	4	X_3	25	2

Cov	X_1	X_2	X_3
X_1	100	50	-10
X_2	50	100	-10
X_3	-10	-10	4

ρ	X_1	X_2	X_3
X_1	1	0.5	-0.5
X_2	0.5	1	-0.5
X_3	-0.5	-0.5	1

What is the variance of the profit $Y = X_1 + 2X_2 + 4X_3$?

$$\mathbf{Var}(Y_1) = \mathbf{Var}(X_1 + 2X_2 + 4X_3) = \mathbf{Cov}(X_1 + 2X_2 + 4X_3, X_1 + 2X_2 + 4X_3)$$

...

$$= \mathbf{Cov}(X_1, X_1) + 4\mathbf{Cov}(X_1, X_2) + 8\mathbf{Cov}(X_1, X_3) + 4\mathbf{Cov}(X_2, X_2) + 16\mathbf{Cov}(X_2, X_3) + 16\mathbf{Cov}(X_3, X_3)$$

$$= 100 + 4 \cdot 50 - 8 \cdot 10 + 4 \cdot 100 - 16 \cdot 10 + 16 \cdot 4 = 524$$

That was tedious! What is the general pattern?

General case: covariance between linear combinations

Let $\boldsymbol{x} = (X_1, X_2, \dots, X_d)$ be random vector with covariances $\mathbf{Cov}(X_i, X_j) = \sigma_{i,j}$ and $a \in \mathbb{R}^d$ and $b \in \mathbb{R}^d$ coefficient vectors. Then covariance between linear combinations $\boldsymbol{a}^T\boldsymbol{x}$ and $\boldsymbol{b}^T\boldsymbol{x}$ is

$$\mathbf{Cov}(\boldsymbol{a}^T\boldsymbol{x}, \boldsymbol{b}^T\boldsymbol{x}) = \sum_{i=1}^{d}\sum_{j=1}^{d} a_i b_j \sigma_{i,j}$$

$$= \sum_{i=1}^{d} a_i b_i \boxed{\sigma_i^2} + 2\sum_{i=1}^{d}\sum_{j=i+1}^{d} a_i b_j \sigma_{i,j}$$

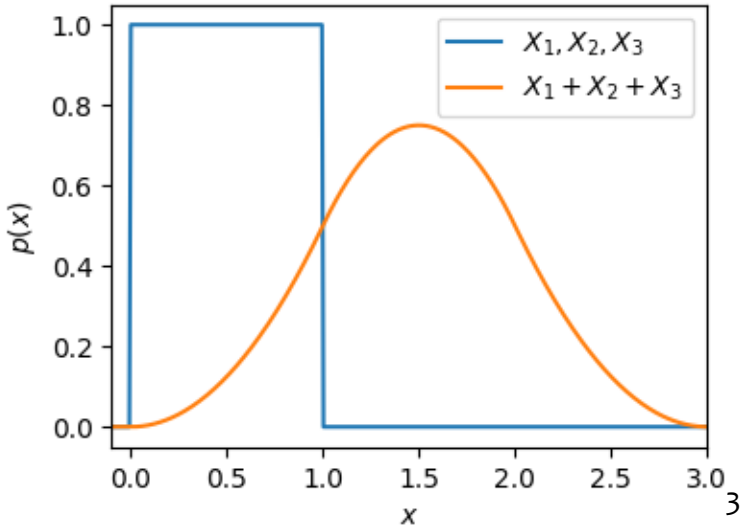
$$= \boldsymbol{a}^T \boldsymbol{\Sigma} \boldsymbol{b}$$

$$\sigma_i^2 = \sigma_{i,i} = \mathbf{Var}(X_i)$$

Special case: $\mathbf{Var}(\boldsymbol{a}^T\boldsymbol{x}) = \boldsymbol{a}^T\boldsymbol{\Sigma}\boldsymbol{a}$

The covariance matrix of $\boldsymbol{x}$ is $d \times d$-matrix $\boldsymbol{\Sigma}$ with entries $\boldsymbol{\Sigma}_{\boldsymbol{i},\boldsymbol{j}} = \sigma_{i,j}$.

$$\boldsymbol{\Sigma} = \begin{bmatrix} \sigma_1^2 & \sigma_{2,1} & \cdots & \sigma_{d,1} \\ \sigma_{1,2} & \sigma_2^2 & & \sigma_{d,2} \\ \vdots & & \ddots & \vdots \\ \sigma_{1,d} & \sigma_{2,d} & \cdots & \sigma_d^2 \end{bmatrix}$$



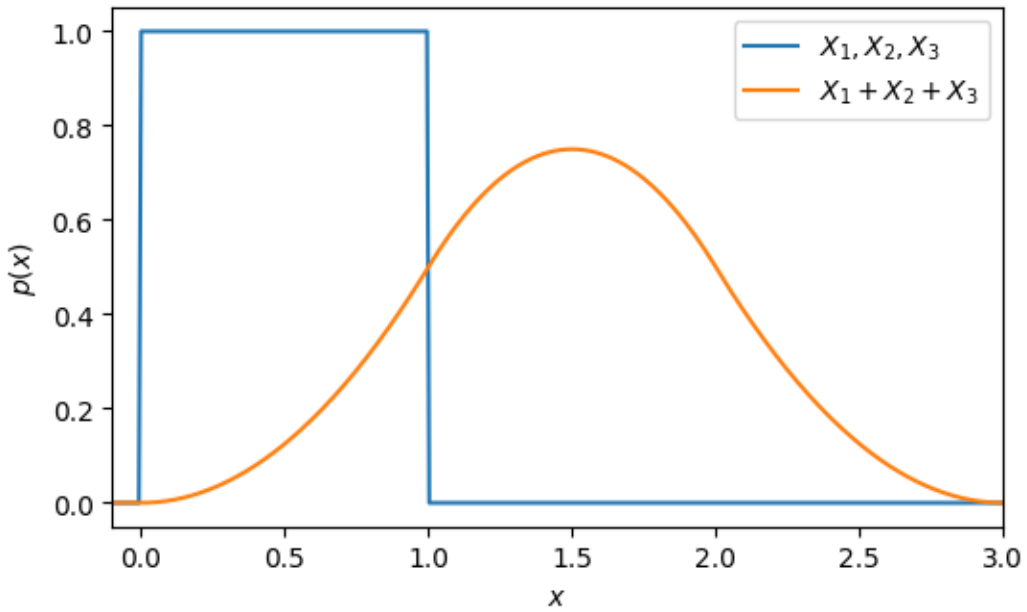
Shape of Linear Combinations of Random Variables

So we can easily compute variance and mean of sums.

What about shape (distribution family)?

In general: shape can change quite rapidly.

Exception: normal distribution!



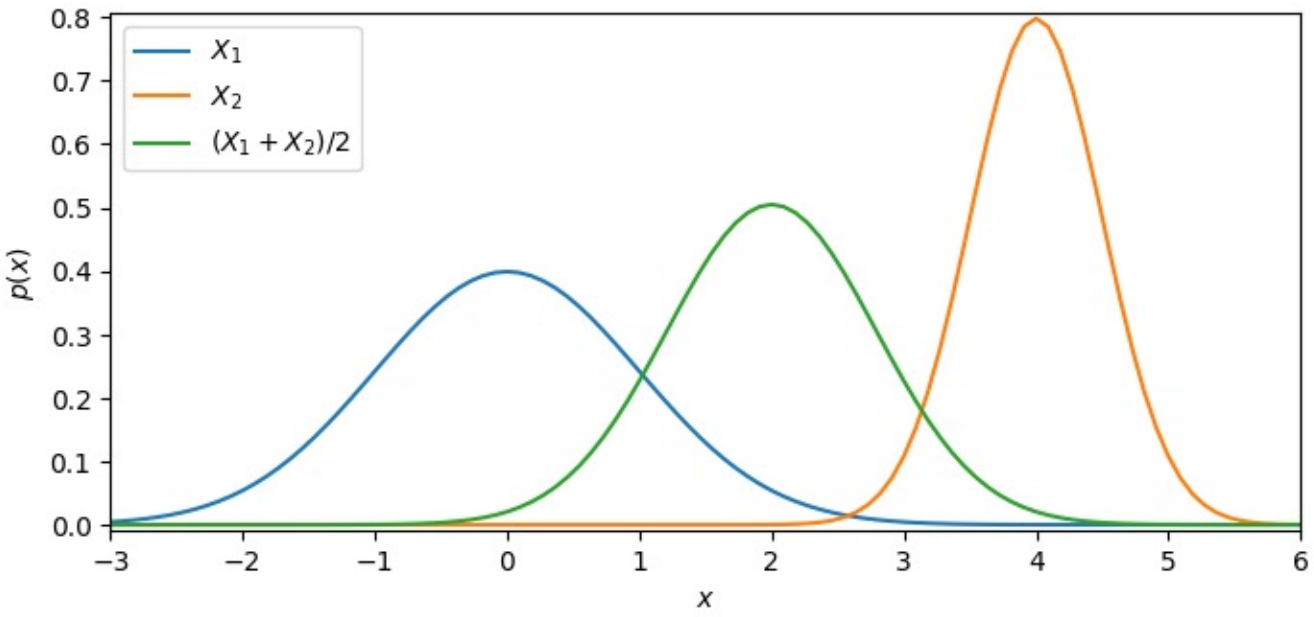
Example: sum of three independent uniform random variables

Linear Combinations of Normal Random Variables

The sum of independent and normally distributed $X_1 \sim \mathrm{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathrm{N}(\mu_2, \sigma_2^2)$ is also normal, i.e.,

$$X_1 + X_2 \sim \mathrm{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

More general: random vector of $\boldsymbol{x} = (X_1, \dots, X_n)$ of independent normal $X_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$ and $\boldsymbol{a} \in \mathbb{R}^n$ has normally distributed linear combination: $\boldsymbol{a}^T \boldsymbol{x} \sim \mathrm{N}(a^T \mu, a_1^2 \sigma_1^2 + \cdots + a_n^2 \sigma_n^2)$



What can we say about linear combinations of dependent normal?

Dependent Normal Random Variables

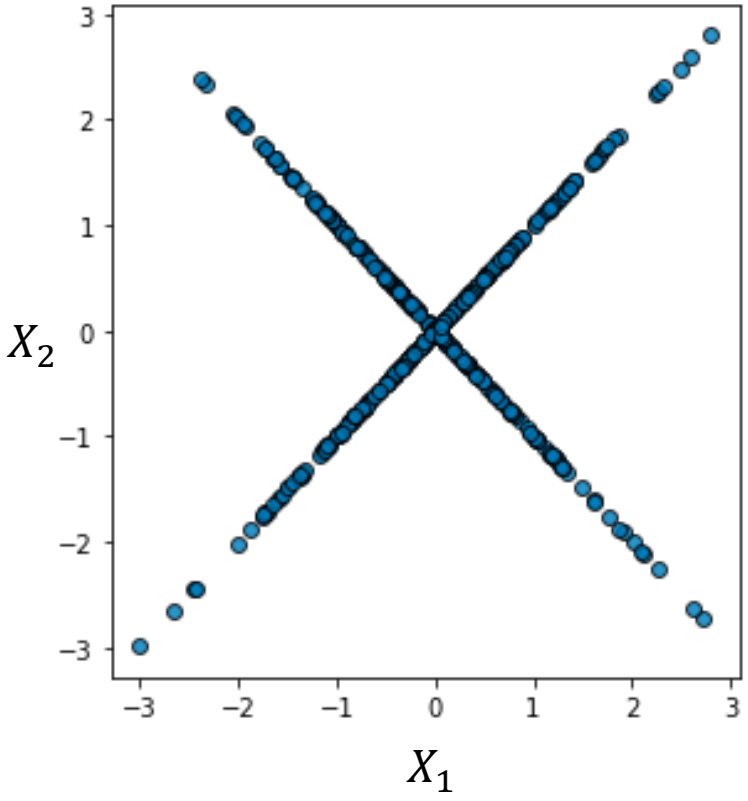
Are sums of dependent normal random variables always normal?

No!

Example: Consider game where, the first player receives/pays X_1 dollars where $X_1 \in \mathrm{N}(0,1)$. Then depending on outcome of coin flip, the second player receives/pays

$$X_2 = \begin{cases} X_1 & \text{if heads} \\ -X_1 & \text{otherwise} \end{cases}$$

With this, marginally, $X_2 \sim \mathrm{N}(0,1)$, but $Z = X_1 + X_2$ is not even a continuous random variable, because $\mathrm{P}(Z = 0) = 0.5$.



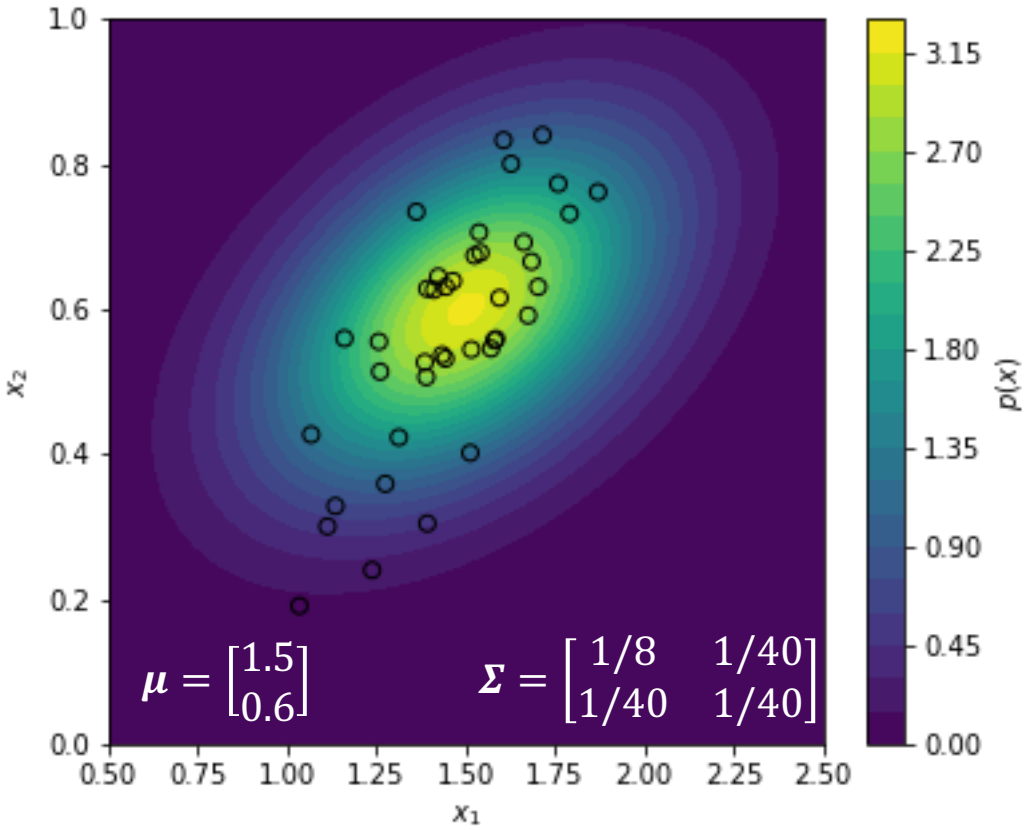
Demanding that all linear combinations of random vector are normal gives rise to multivariate normal distribution!

Multivariate Normal Distribution

Random vector $\boldsymbol{x} = (X_1, \dots, X_d)$ with mean vector $\boldsymbol{\mu}$ and covariance matrix $\boldsymbol{\Sigma}$ follows multivariate normal distribution if for all coefficient vectors the linear combination is normal:

$$\boldsymbol{a}^T \boldsymbol{x} \sim \mathrm{N}(\boldsymbol{a}^T \boldsymbol{\mu}, \boldsymbol{a}^T \boldsymbol{\Sigma} \boldsymbol{a})$$

This is denoted by $\boldsymbol{x} \sim \mathrm{N}_d(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.



The joint probability density function is given by:

$$p(\boldsymbol{x}) = \frac{1}{\sqrt{(2\pi)^d \boxed{|\boldsymbol{\Sigma}|}}} \exp\left(-\frac{1}{2} \boxed{(\boldsymbol{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1} (\boldsymbol{x} - \boldsymbol{\mu})}\right)$$

determinant (deformation of unit hypercube by matrix)

squared length of standardised version of value

generalises univariate case $\quad p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \boxed{\frac{(x-\mu)^2}{\sigma^2}}\right)$

Probabilistic Generative Models with Multiple Input Variables

How to Handle Multiple Input Variables

Assume again multiple (continuous) input variables: $\boldsymbol{X} = (X_1, \dots, X_d) \in \mathbb{R}^d$.

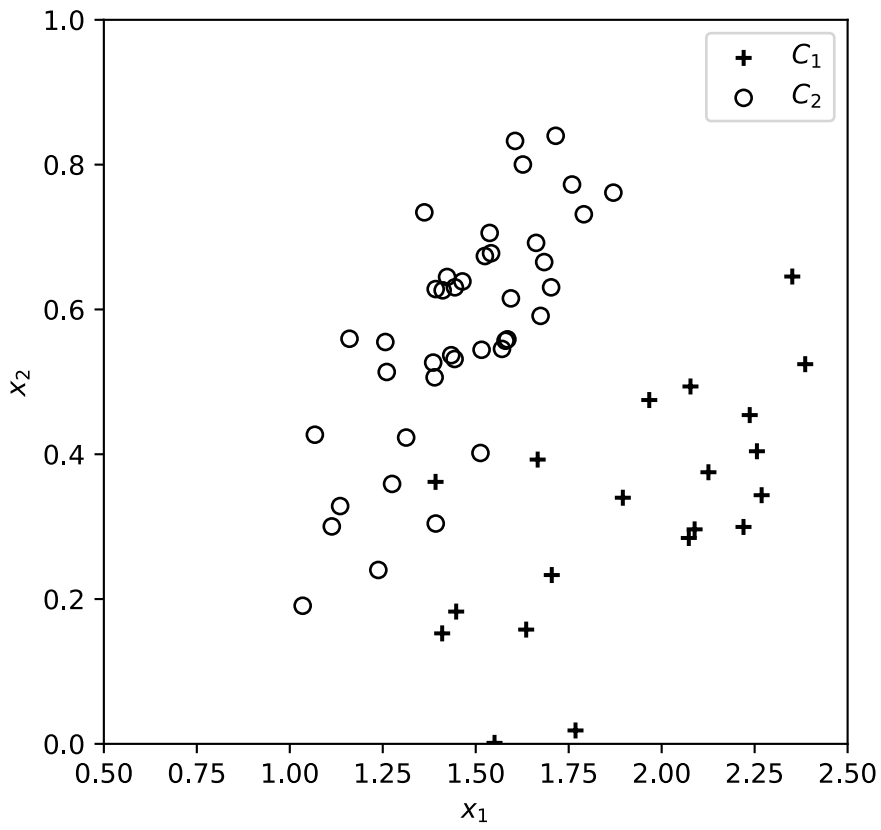
How to model class conditional input densities?

Idea 1: assume conditionally independent inputs

$$p(\boldsymbol{x}|c_k) = p(x_1|c_k)p(x_2|c_k) \cdots p(x_d|c_k)$$

-> allows to re-use previous univariate Gaussians
-> approach is called Naïve Bayes
-> however, independence assumption often violated

Idea 2: model as multivariate normal distribution



Parameter Fitting for Multivariate Normal Class Conditionals

Per class need to fit mean vector $\boldsymbol{\mu}_k$ and covariance matrix $\boldsymbol{\Sigma}_k$

$$p(\boldsymbol{x}|c_k) = \frac{1}{\sqrt{(2\pi)^d|\boldsymbol{\Sigma}_k|}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}_k^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k)\right)$$

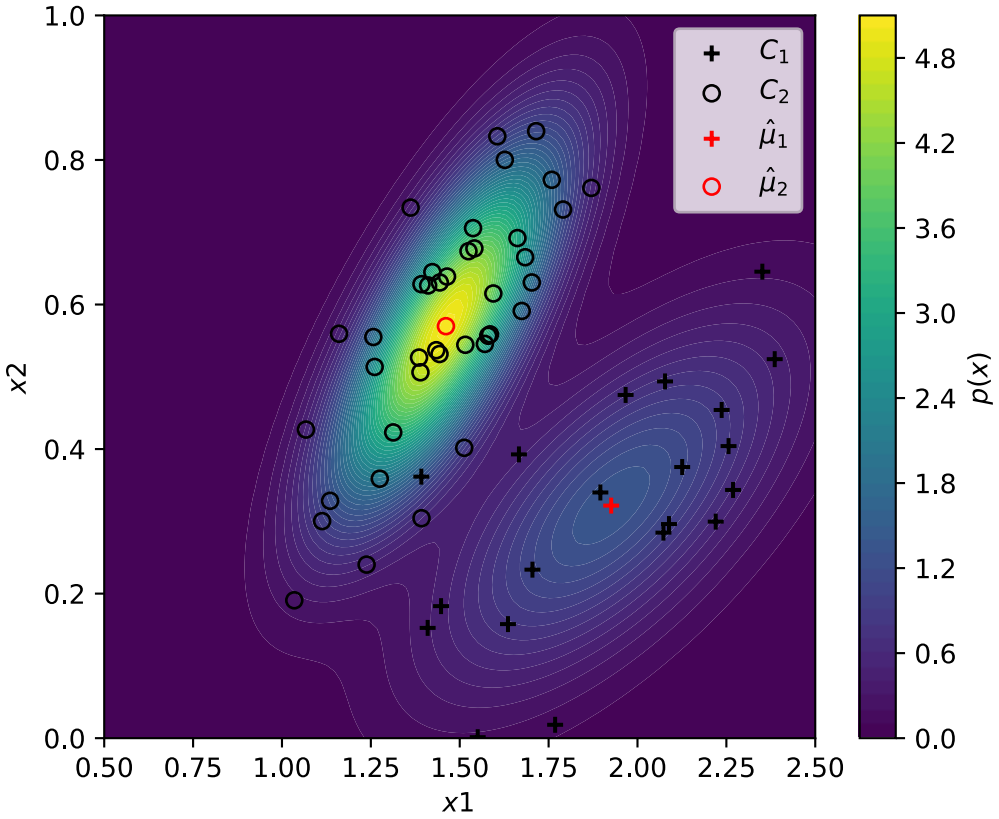
Split again data via classes and find ML estimates as

$$\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n\in D_k}\boldsymbol{x}_n$$

$$\boldsymbol{\Sigma}_k = \frac{1}{N_k}\sum_{n\in D_k}(\boldsymbol{x}_n-\boldsymbol{\mu}_k)(\boldsymbol{x}_n-\boldsymbol{\mu}_k)^T$$

or component-wise

$$\sigma_{i,j} = \frac{1}{N_k}\sum_{n\in D_k}(x_{n,i}-\mu_{k,i})(x_{n,j}-\mu_{k,j})$$



Total of Kd^2 covariance parameters to fit!
Danger of overfitting if $n \approx Kd^2$.

Reduced Parameter Variant 1: Diagonal Covariance Matrices

Per class need to fit mean vector $\boldsymbol{\mu}_k$ and covariance matrix Σ_k

$$p(\boldsymbol{x}|c_k) = \frac{1}{\sqrt{(2\pi)^d|\boldsymbol{\Sigma}_k|}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^T\Sigma_k^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k)\right)$$

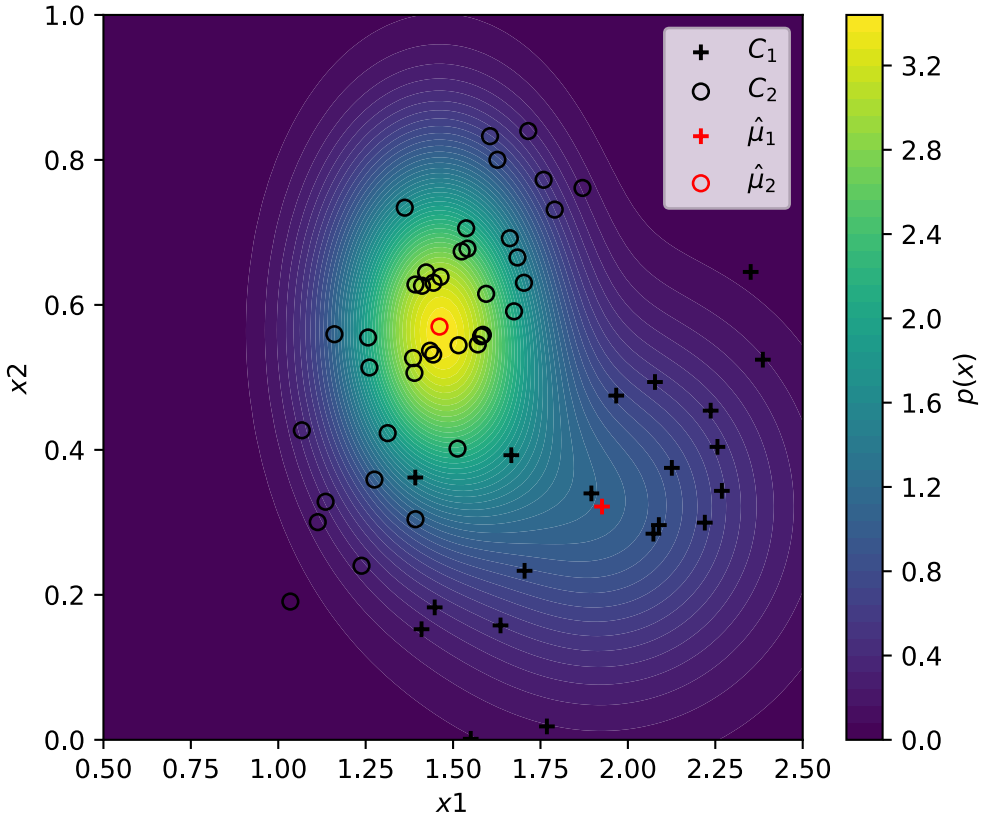
Split again data via classes and find ML estimates as

$$\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n\in D_k}\boldsymbol{x}_n$$

$$\boldsymbol{\Sigma}_k = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_d^2 \end{bmatrix}$$

with components

$$\sigma_i^2 = \frac{1}{N_k}\sum_{n\in D_k}\left(x_{n,i}-\mu_{k,i}\right)^2$$



Only of Kd variance parameters.
This is just Naïve Bayes! Might not fit data well.

Reduced Parameter Variant 2: Shared Covariance Matrix

Per class need to fit mean vector $\boldsymbol{\mu}_k$ and covariance matrix Σ_k

$$p(\boldsymbol{x}|c_k) = \frac{1}{\sqrt{(2\pi)^d|\boldsymbol{\Sigma}|}}\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_k)^T\boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_k)\right)$$

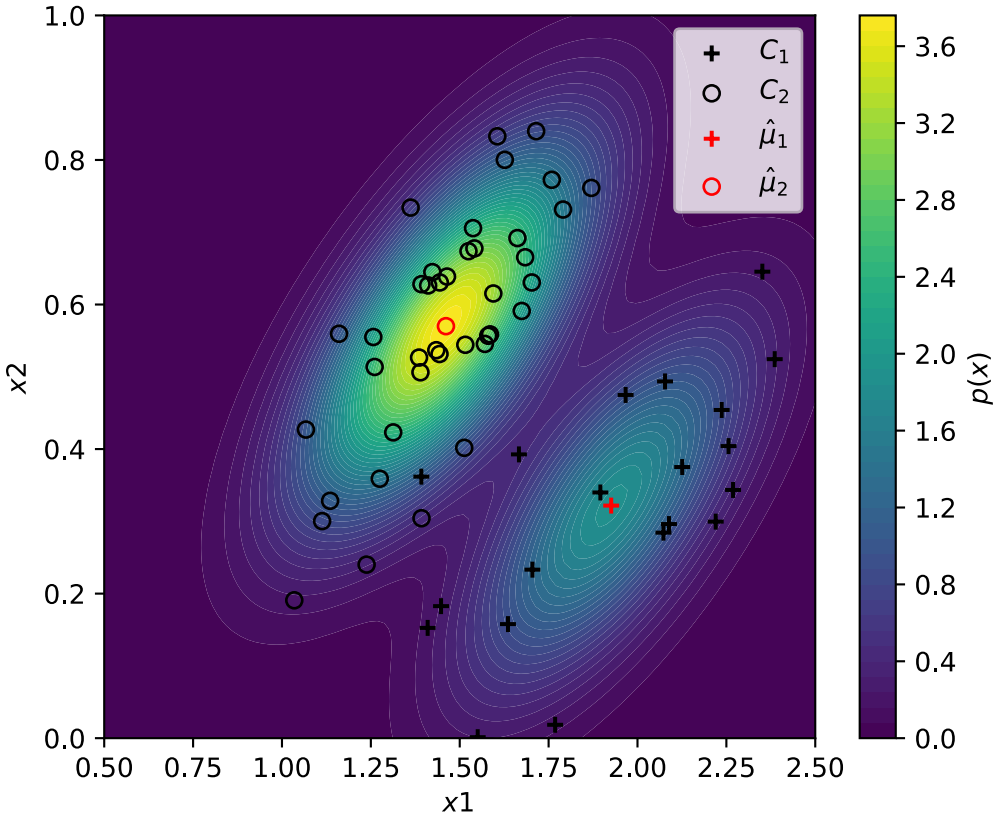
Split again data via classes and find ML estimates as

$$\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_{n\in D_k}\boldsymbol{x}_n$$

$$\Sigma_k = \frac{1}{N_k}\sum_{n\in D_k}(\boldsymbol{x}_n-\boldsymbol{\mu}_k)(\boldsymbol{x}_n-\boldsymbol{\mu}_k)^T$$

or component-wise

$$\sigma_{i,j} = \frac{1}{N_k}\sum_{n\in D_k}(x_{n,i}-\mu_{k,i})(x_{n,j}-\mu_{k,j})$$



$$\boldsymbol{\Sigma} = \sum_{k=1}^{K}\frac{N_K}{N}\Sigma_k$$

Middle ground: single covariance matrix

Still d^2 covariance parameters

Decision Boundary

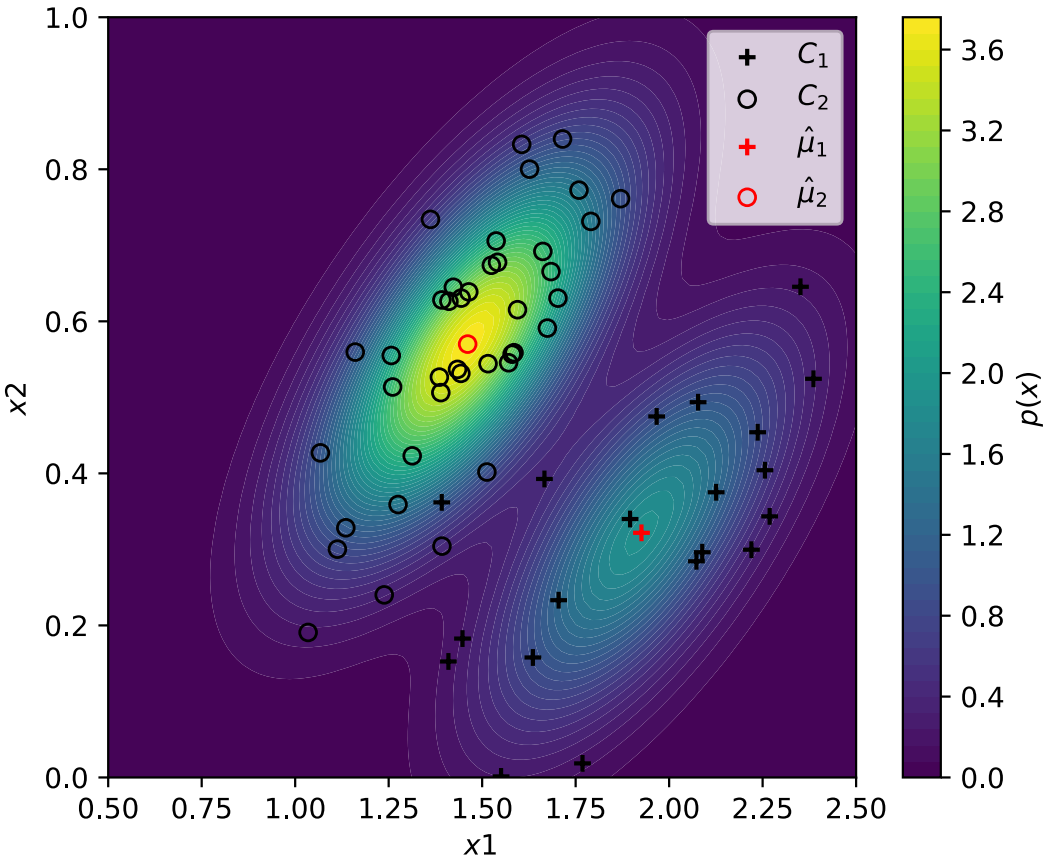
Can use model for prediction with uncertainties, but can we describe decision boundary algebraically as before?

$$p(x|c_1)p(c_1) = p(x|c_2)p(c_2)$$

Alternatively: consider log odds.

$$a = \ln \frac{p(x|c_1)p(c_1)}{p(x|c_2)p(c_2)} = 0$$

$$= \ln \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_1)^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1)\right)}{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_2)^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_2)\right)} + \ln \frac{p(c_1)}{p(c_2)}$$



Is this still a linear model in the previous sense?

Decision Boundary

$$a = \ln \frac{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_1)^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1)\right)}{\exp\left(-\frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_2)^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_2)\right)} + \ln\frac{p(c_1)}{p(c_2)}$$

$$= \frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_2)^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_2) - \frac{1}{2}(\boldsymbol{x}-\boldsymbol{\mu}_1)^T\Sigma^{-1}(\boldsymbol{x}-\boldsymbol{\mu}_1) + \ln\frac{p(c_1)}{p(c_2)}$$

quadratic terms cancel because of assumption of single covariance matrix

$$= \cancel{\frac{1}{2}\boldsymbol{x}^T\Sigma^{-1}\boldsymbol{x}} - \boldsymbol{x}^T\Sigma^{-1}\boldsymbol{\mu}_2 + \frac{1}{2}\boldsymbol{\mu}_2\Sigma^{-1}\boldsymbol{\mu}_2 - \cancel{\frac{1}{2}\boldsymbol{x}^T\Sigma^{-1}\boldsymbol{x}} + \boldsymbol{x}^T\Sigma^{-1}\boldsymbol{\mu}_1 - \frac{1}{2}\boldsymbol{\mu}_1\Sigma^{-1}\boldsymbol{\mu}_1 + \ln\frac{p(c_1)}{p(c_2)}$$

$$= \boxed{(\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)^T\Sigma^{-1}}\boldsymbol{x} + \boxed{\frac{1}{2}\boldsymbol{\mu}_2^T\Sigma^{-1}\boldsymbol{\mu}_2 - \frac{1}{2}\boldsymbol{\mu}_1^T\Sigma^{-1}\boldsymbol{\mu}_1 + \ln\frac{p(c_1)}{p(c_2)}}$$

$$a = \boldsymbol{w}^T\boldsymbol{x} + w_0$$

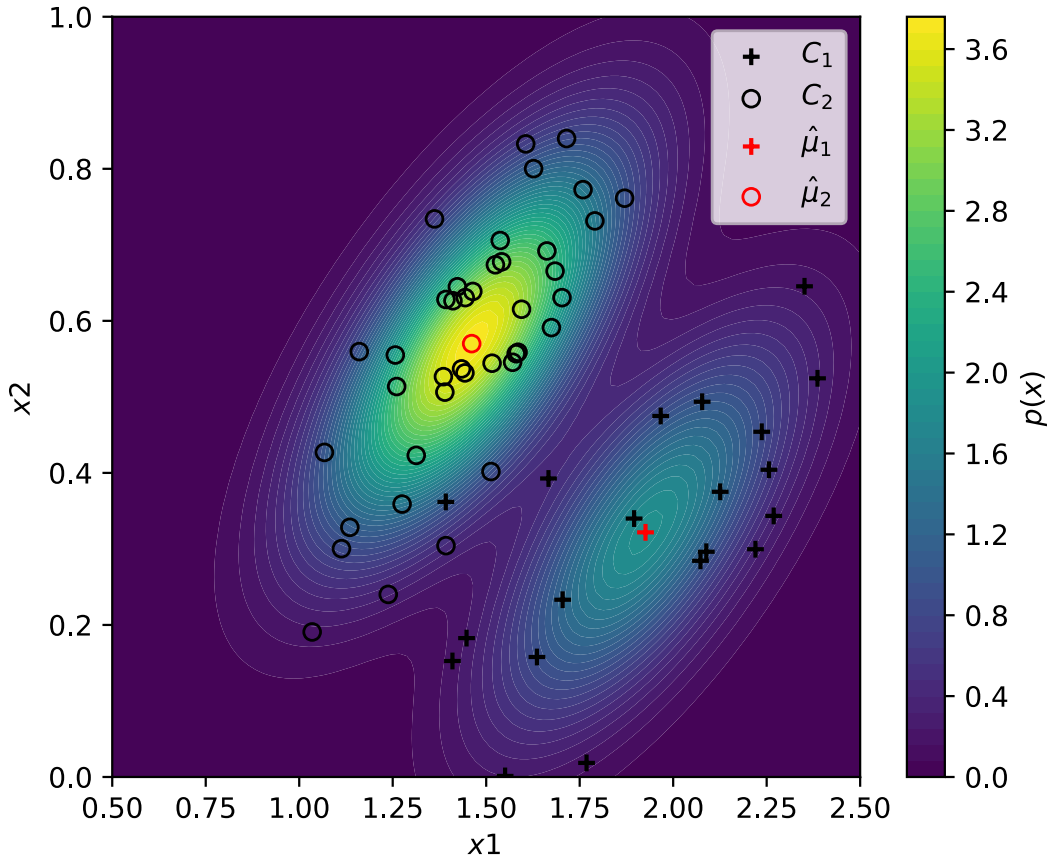
log odds again described by linear function of weights:

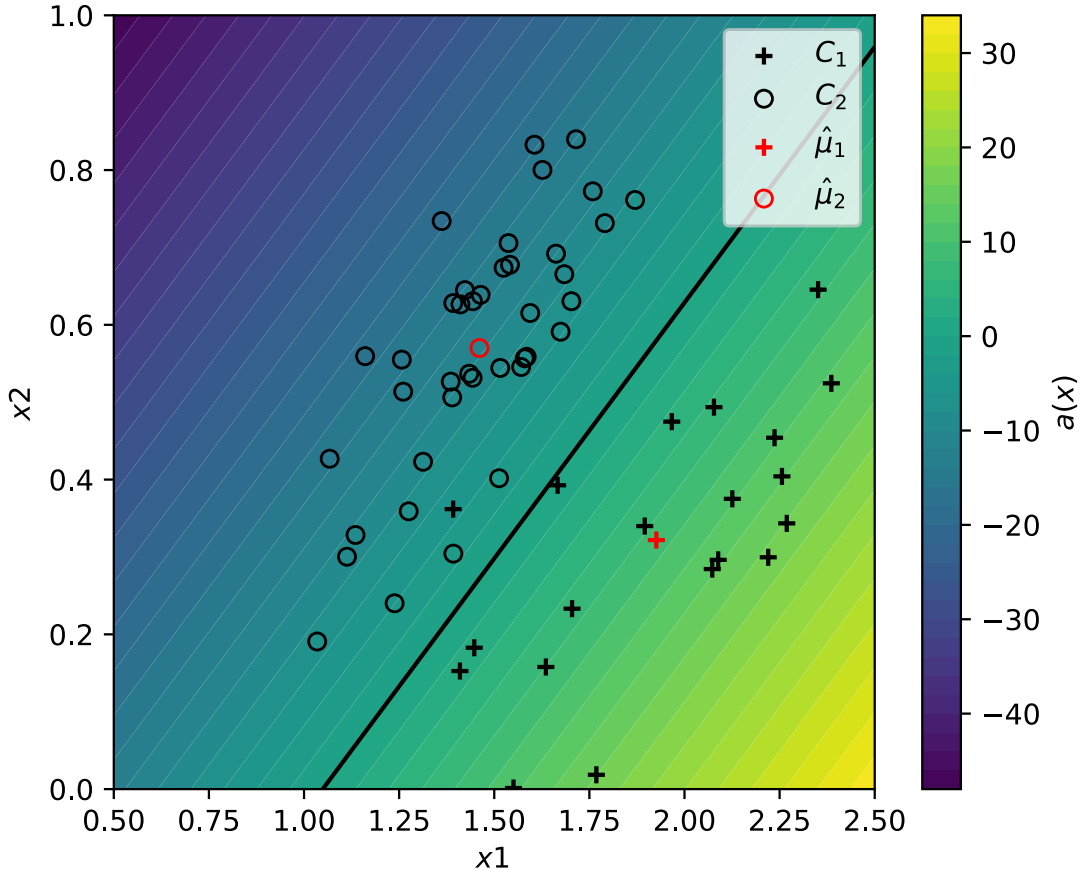
$$\boldsymbol{w} = (\boldsymbol{\mu}_1-\boldsymbol{\mu}_2)^T\Sigma^{-1}$$

$$w_0 = \frac{1}{2}\boldsymbol{\mu}_2^T\Sigma^{-1}\boldsymbol{\mu}_2 - \frac{1}{2}\boldsymbol{\mu}_1^T\Sigma^{-1}\boldsymbol{\mu}_1 + \ln\frac{p(c_1)}{p(c_2)}$$

Single Covariance Matrix implies Linear Log Odds

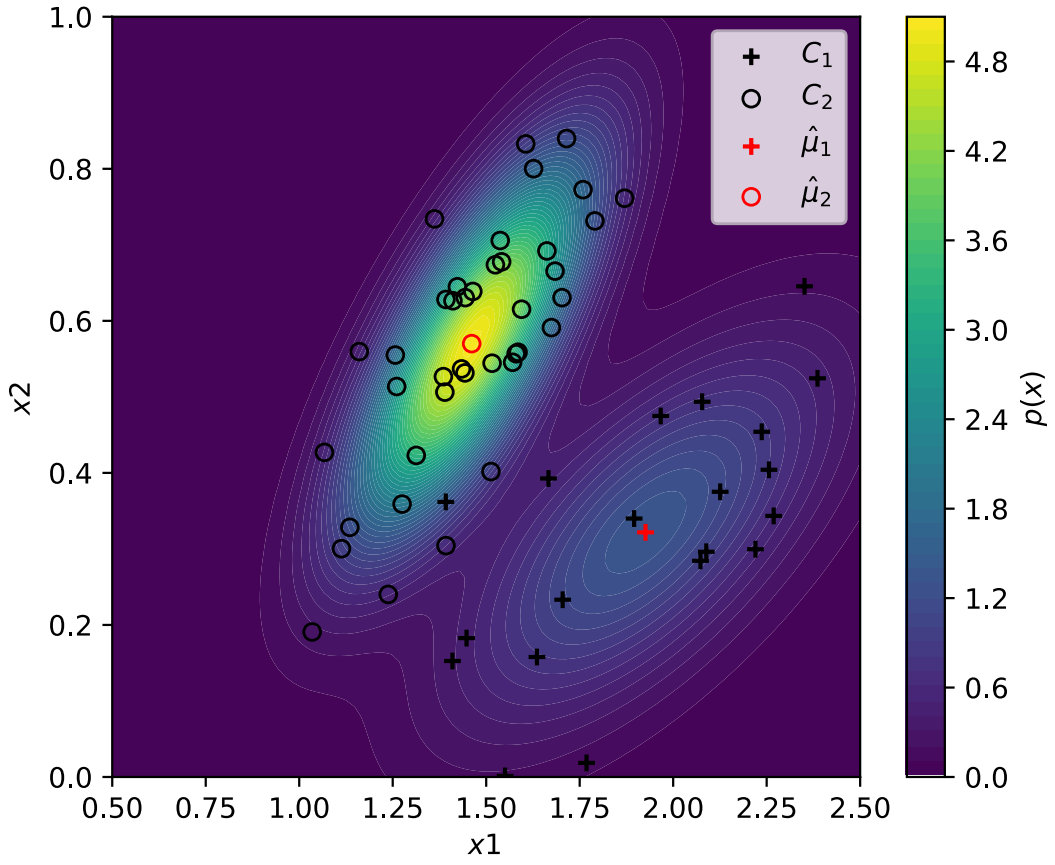
$$a = \boldsymbol{w}^T \boldsymbol{x} + w_0 \qquad \text{with} \qquad \boldsymbol{w} = (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^T \Sigma^{-1} \qquad w_0 = \frac{1}{2}\boldsymbol{\mu}_2^{\boldsymbol{T}}\Sigma^{-1}\boldsymbol{\mu}_2 - \frac{1}{2}\boldsymbol{\mu}_1^{\boldsymbol{T}}\Sigma^{-1}\boldsymbol{\mu}_1 + \ln\frac{p(c_1)}{p(c_2)}$$

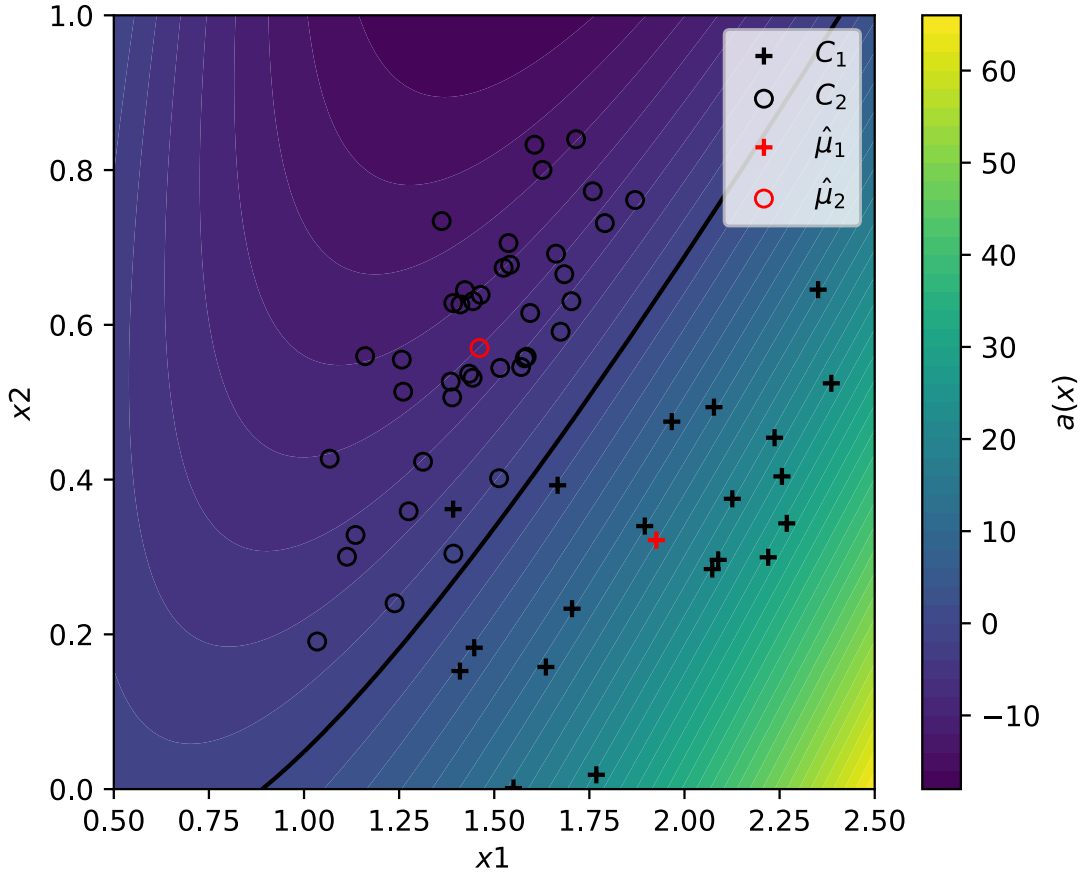




Class-specific Covariances lead to Quadratic Log Odds

$$a = \boldsymbol{x}^T A \boldsymbol{x} + \boldsymbol{w}^T \boldsymbol{x} + w_0 \qquad \text{with} \qquad A = \frac{1}{2}(\Sigma_2^{-1} - \Sigma_1^{-1}) \qquad \boldsymbol{w} = \boldsymbol{\mu}_1^T \Sigma_1^{-1} - \boldsymbol{\mu}_2^T \Sigma_2^{-1}$$





Generative Models: Concluding Remarks

- Generative models (Bayes classifier) with normal class conditionals densities are attractive option because they allow to find ML parameters in closed form without expensive iterative algorithm

- However, they require to choose conditional input distributions (and sometimes multivariate normal might not capture the data well)

- Also, if not using Naïve Bayes assumption, there are relatively many parameters to fit (quadratic order of number of input features)

- Can use shared covariance to reduce parameters by factor K

- Can be extended to discrete features, typically in combination with the Naïve Bayes assumption, where each feature is modelled class-conditionally as categorical random variable

THANK YOU
for your attention!